

ASSIGNMENT NO: 01

Course Title : Matrices, Complex Variable and
Fourier Analysis

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Answer of question:01

SUPPOSE,

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\begin{aligned} \therefore (A) &= 1(3-0) - 2(-1-0) - 2(2-0) \\ &= 3 + 2 - 4 = 1 \neq 0 \end{aligned}$$

Hence A^{-1} exists

Now we need to determine coefficients:

$$a_{11} = 3 - 0 = 3$$

$$a_{12} = -(-1-0) = 1$$

$$a_{13} = 2 - 0 = 2$$

$$a_{21} = -(2-4) = 2$$

$$a_{22} = (1-0) = 1$$

$$a_{23} = -(-2-0) = 2$$

$$a_{31} = 0 - (-6) = 6$$

$$a_{32} = -(0-2) = 2$$

$$a_{33} = 3 - (-2) = 5$$

$$\therefore \text{cof}(A) = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 1 & 2 \\ 6 & 2 & 5 \end{bmatrix}$$

$$\therefore \text{adj}(A) = \text{cof}(A)^T = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

$$\begin{aligned} \therefore A^{-1} &= \frac{1}{|A|} \text{adj}(A) \\ &= \frac{1}{1} \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} \end{aligned}$$

Therefore, inverse of A is $\begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$

Ans

Answer of Question: 02

∴ Gauss - Elimination:

From the given system,

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 0 \end{bmatrix}$$

Augmented form:

$$\begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 2 & 3 & 10 \\ 2 & 10 & 1 & 6 \end{bmatrix} \quad \begin{array}{l} \text{substituting} \\ r_1 \text{ and } r_3 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 2 & 10 \\ 2 & 10 & 1 & 6 \end{bmatrix} \quad r'_2 = r_2 - r_1$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 10 & 1 & 6 \\ 0 & 0 & 2 & 10 \end{array} \right] \text{ substituting } r_2 \text{ and } r_3$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 6 & -1 & 6 \\ 0 & 0 & 2 & 10 \end{array} \right] r'_2 = r_2 - 2r_1$$

By back substitution,

$$2z = 10 \quad \therefore z = 5$$

$$6y - z = 6$$

$$\Rightarrow 6y = 6 + z$$

$$\Rightarrow y = \frac{6+5}{6} = \frac{11}{6} = y$$

$$x + 2y + z = 0$$

$$\Rightarrow x = -z - 2y = -5 - 2 \cdot \frac{11}{6} = -5 - \frac{11}{3}$$

$$= \frac{-15-11}{3} = \frac{-26}{3} = x$$

$$\therefore (x, y, z) = \left(\frac{-26}{3}, \frac{11}{6}, 5 \right) \underline{\underline{\text{Ans.}}}$$

Answer of Question: 03

Suppose,

$$A = \begin{bmatrix} -5 & 12 \\ 0 & -2, 0 \\ 4 & 2 -3 \end{bmatrix}$$

$$\therefore \lambda I - A = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -5 & 12 \\ 0 & -2 & 0 \\ 4 & 2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} -5 & 12 \\ 0 & -2 & 0 \\ 4 & 2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda+5 & -12 & -2 \\ 0 & \lambda+2 & 0 \\ -4 & -2 & \lambda+3 \end{bmatrix}$$

$$\therefore |\lambda I - A| = (\lambda+5)(\lambda^2+3\lambda+2\lambda+6) + 0 - 2(0+4\lambda+8)$$

$$= (\lambda+5)(\lambda^2+5\lambda+6) - 8\lambda - 16$$

$$= \lambda^2 + 5\lambda^2 + 6\lambda + 5\lambda^2 + 25\lambda + 30 - 80\lambda - 16$$

$$= \lambda^2 + 20\lambda^2 + 23\lambda + 14$$

characteristics equation:

$$\lambda^3 + 10\lambda^2 + 23\lambda + 14 = 0$$

$$\therefore \lambda_1 = -7, \lambda_2 = -2, \lambda_3 = -2$$

(i) When, $\lambda = -7$

$$(\lambda I - A)x = \begin{bmatrix} -7+5 & -1 & -2 \\ 0 & -7+2 & 0 \\ -4 & -2 & -7+3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2 & -1 & -2 \\ 0 & -5 & 0 \\ -4 & -2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2x_1 - x_2 - 2x_3 \\ 0 - 5x_2 + 0 \\ -4x_1 - 2x_2 - 4x_3 \end{bmatrix} = 0$$

$$\therefore (i) \rightarrow -2x_1 - x_2 - 2x_3 = 0 \Rightarrow 2x_1 + x_2 + 2x_3 = 0$$

$$(ii) \rightarrow -5x_2 = 0 \Rightarrow 5x_2 = 0 \therefore x_2 = 0$$

$$(iii) \rightarrow -4x_1 - 2x_2 - 4x_3 = 0 \Rightarrow 4x_1 + 2x_2 + 4x_3 = 0$$

$$\Rightarrow 4x_1 + 4x_3 = 0 \Rightarrow (x_1 + x_3) = 0$$

$$\Rightarrow x_1 = -x_3$$

$$(i) \rightarrow 2x_1 + 2x_3 = 0 \Rightarrow x_1 = -x_3$$

NOW, we know

$$x_1 + x_2 + x_3 = 1$$

$$\Rightarrow (-x_3) + 0 + x_3 = 1$$

$$\Rightarrow x_3 + x_3 = 1$$

$$\therefore x_3 = \frac{1}{\sqrt{2}}$$

$$\therefore x_1 = -\frac{1}{\sqrt{2}}$$

$$\therefore \text{eigenvector, } x = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

(ii) when, $\lambda = -1$

$$(\lambda I - A)x = \begin{bmatrix} -1+5 & -1 & -2 \\ 0 & -1+2 & 0 \\ -4 & -2 & -1+3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 4x_1 - x_2 - 2x_3 \\ 0 + x_2 + 0 \\ -4x_1 - 2x_2 + 2x_3 \end{bmatrix} = 0$$

$$(i) \rightarrow 4x_1 - x_2 - 2x_3 = 0$$

$$(ii) \rightarrow x_2 = 0$$

$$(iii) \rightarrow 4x_1 - 0 + 2x_3 = 0 \Rightarrow 2(2x_1 + x_3) = 0 \\ \Rightarrow x_3 = -2x_1$$

We know,

$$x_1^2 + x_2^2 + x_3^2 = 1$$

$$\Rightarrow x_1^2 + 0 + (-2x_1)^2 = 1$$

$$\Rightarrow x_1^2 + 4x_1^2 = 1 \Rightarrow x_1 = \frac{1}{\sqrt{5}}$$

$$\therefore x_3 = -2 \frac{1}{\sqrt{5}}$$

$\therefore$ eigen vector, $x = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 0 \\ \frac{-2}{\sqrt{5}} \end{bmatrix}$

(iii) When $\lambda = -2$

$$(\lambda I - A)x = \begin{bmatrix} -2+5 & -1 & -2 \\ 0 & -2+2 & 0 \\ -4 & -2 & -2+3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 3x_1 - x_2 - 2x_3 \\ 0 + 0 + 0 \\ -4x_1 - 2x_2 + x_3 \end{bmatrix} = 0$$

We have 3 variables, but 2 equations, such system cannot be solved.

So there is no any defined eigenvector for $\lambda = -2$.

Ans

Answer of question : 04

If we consider a linear equations system,

$$Ax = B,$$

① consistent system:

A system is consistent when it has at least 1 solution.

condition:

$$\text{Rank}(A) = \text{Rank}([A|B])$$

① if, $\text{Rank}(A) = \text{Rank}([A|B]) = n \rightarrow$ unique solution

② if, $\text{Rank}(A) = \text{Rank}([A|B]) < n \rightarrow$ infinitely many solutions.

② Inconsistent system:

A system is inconsistent if it has no solution.

condition:

$$\text{Rank}(A) \neq \text{Rank}([A|B])$$

for homogeneous system:

A homogeneous system never has no trivial solution.

It is always consistent because the trivial solution,

$$x=0$$

Ans:

Answer of Question: 05

According to de Moivre's theorem,

$$\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$$

$$\Rightarrow \cos 4\theta + i \sin 4\theta = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4 \cos \theta i \sin^3 \theta + i^4 \sin^4 \theta$$

$$\Rightarrow \cos 4\theta + i \sin 4\theta = \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$

$$\Rightarrow \cos 4\theta + i \sin 4\theta = (\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) + i (4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta)$$

Equating real and imaginary part,

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\Rightarrow \cos^4 \theta = \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$\Rightarrow \cos^4 \theta = \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$\Rightarrow \cos^4 \theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$$

$$\Rightarrow 8 \cos^4 \theta = \cos^4 \theta + 8 \cos^2 \theta - 1$$

$$\Rightarrow \cos^4 \theta = \frac{1}{8} (\cos^4 \theta + 2 \cos 2 \theta + 3) \text{ (Proved)}$$

Answer of Question 866)

Here,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\therefore e^{-i\theta} = \cos \theta + (-i) \sin \theta \\ = \cos \theta - i \sin \theta$$

$$\text{Now, R.H.S} = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$= \frac{\cos \theta + i \sin \theta - (\cos \theta - i \sin \theta)}{2i}$$

$$= \frac{\cos \theta + i \sin \theta - \cos \theta + i \sin \theta}{2i}$$

$$= \frac{2i \sin \theta}{2i}$$

$$= 2 \sin \theta = \text{L.H.S} \text{ (Proved)}$$

Answer of Question: 07

Hence,

$$z^4 = 2 + i\sqrt{3} \Rightarrow z = (2 + i\sqrt{3})^{1/4}$$

Now, for $(2 + i\sqrt{3})$,

$$r = \sqrt{2^2 + (\sqrt{3})^2} = \sqrt{4+3} = 2$$

$$\theta = \tan^{-1}\sqrt{3} = \frac{\pi}{3}$$

$$\therefore (2 + i\sqrt{3})^{1/4} = \left[2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \right]^{1/4}$$

$$= 2^{1/4} \left[\cos \left(2k\pi + \frac{\pi}{3} \right) + i \sin \left(2k\pi + \frac{\pi}{3} \right) \right]^{1/4}$$

$$= 2^{1/4} \left[\cos \frac{6k\pi + \pi}{3} + i \sin \frac{6k\pi + \pi}{3} \right]^{1/4}$$

$$= 2^{1/4} \left[\cos \frac{6k\pi + \pi}{12} + i \sin \frac{6k\pi + \pi}{12} \right]$$

when, $k=0$,

$$z_1 = 2^{1/4} \left[\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right]$$

when, $k=1$,

$$z_2 = 2^{1/4} \left[\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right]$$

when, $k=2$,

$$z_3 = 2^{1/4} \left[\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right]$$

when, $k=3$,

$$Z_4 = 2^{1/4} \left[e^{i \cos \frac{12\pi}{12}} + i \sin \frac{12\pi}{12} \right]$$

Ans 0

Answer of question 2(08)

② given,

$$f(x) = x + x^2; -\pi \leq x \leq \pi$$

we know,

$$a_0 = \frac{2}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{\pi} x dx + \int_{-\pi}^{\pi} x^2 dx \right]$$

$$= \frac{1}{\pi} \left[\frac{x^2}{2} \right]_{-\pi}^{\pi} + \frac{x^3}{3} \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{1}{2} [\pi^2 - (-\pi^2)] + \frac{1}{3} [\pi^3 - (-\pi^3)] \right]$$

$$= \frac{1}{\pi} \left[+ \left(\frac{1}{3} \times 2\pi^3 \right) \right]$$

$$= \frac{1}{\pi} \left(+ \frac{2}{3} \pi^3 \right) = \frac{2}{3} \pi^2 = a_0$$

Now,

$$a_n = \frac{1}{\pi} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x+x^2) \cos \frac{n\pi x}{\pi} dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x+x^2) \cos nx dx$$

$$= \frac{1}{\pi} \left[(x+x^2) \left(+ \frac{\sin nx}{n} \right) - (1+2x) \left(- \frac{\cos nx}{n^2} + (0+2) \left(- \frac{\sin nx}{n^3} \right) \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(\pi + \pi^2) \frac{0}{n} + (1+2\pi) \left(\frac{\cos n\pi}{n^2} \right) - 2 \frac{0}{n^3} - \left\{ (-\pi + \pi^2) \frac{0}{n} + (1-2\pi) \cdot \frac{\cos n\pi}{n^2} + 2 \frac{0}{n^3} \right\} \right]$$

$$= \frac{1}{\pi} \left(0 + \left(\frac{1+2\pi}{n^2} \right) \cdot (-1) - 0 - 0 - \left(\frac{1-2\pi}{n^2} \right) \cdot (-1) + 0 \right)$$

$$= \frac{4}{n^2} (-1)^n = a_n$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (x+x^2) \sin nx dx$$

$$= \frac{1}{\pi} \left[(x+x^2) \left(- \frac{\cos nx}{n} \right) - (1+2x) \left(\frac{\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(\pi + \pi^2) \left(-\frac{\cos n\pi}{n} \right) + (1 + 2\pi) \cdot \frac{0}{n^2} + \frac{2\cos n\pi}{n^3} \right. \\ \left. - \left\{ -\pi - \pi^2 \right\} \cdot \left(-\frac{\cos n\pi}{n} \right) + (1 - 2\pi) \frac{0}{n^2} + 2 \frac{\cos n\pi}{n^3} \right]$$

$$= \frac{1}{\pi} \left[\left(\frac{\cos n\pi}{n} \right) (\pi + \cancel{\pi^2} + \pi - \cancel{\pi^2}) + 2 \frac{\cos n\pi}{n^3} - \frac{\cos n\pi}{n^3} \right]$$

$$= \frac{1}{\pi} \frac{-2\pi(-1)^n}{n} = \frac{-2(-1)^3}{n} = b_n$$

(ii) Therefore Fourier Series is :

$$f(x) = \frac{2\pi^2}{3} \cdot \frac{1}{2} + \sum \left[\frac{4(-1)^3}{n^2} \cos nx + \left(-\frac{2(1)^3}{n} \right) \sin nx \right]$$

$$= \frac{\pi^2}{3} + 4 \sum \frac{(-1)^3}{n^2} \cos nx = 2 \sum \frac{(1)^3}{n} \sin nx$$

$$= \frac{\pi^2}{3} + 4 \sum \left(-\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \dots \right) - 2 \left(-\frac{\sin x}{1} + \frac{\sin 2x}{2} - \dots \right)$$

$$= \frac{\pi^2}{3} - 4 \left(\cos x - \cos \frac{2x}{2^2} + \dots \right) + 2 \sin x - \frac{\sin 2x}{2} + \dots$$

Ans :